

Assignment 5: Test Driven Development Software Engineering for Scientists

Objectives: Practice test driven development and combine two data sets that share a common field.

Tasks:

1. Fork the repo <https://github.com/swe4s/tdd>
2. Clone **your** fork (NOT <https://github.com/swe4s/tdd>)
3. Download the two data files in the README and put them in a folder named data at the top of your repo
 - a. **Do not commit these files.** This is made easier by putting them into the data folder, which is already in .gitignore.
 - b. Agrofood_co2_emission.csv has one row per country per year from 1990 to 2020. The country is in the Area column (index 0) and the year is in the Year column (index 1). Fire emissions are in the Forest fires column. Some cells are empty because the value is missing.
 - c. IMF_GDP.csv has one row per country. The country is in the Country column (index 0), and each remaining column is a year from 1950 to 2022. Some cells are empty because the value is missing. GDP is in millions of each country's own currency, so values from different countries are not directly comparable.
 - d. Create small test files and put them in test/data. **Commit these files** and use them in your testing and development.
4. Using test driven development, write the three functions in src/fire_gdp.py.
 - a. get_data(file_name, query_column=None, query_value=None, return_header=False) returns rows from a data file. If a query column and value are given, return only the rows where the string in that column matches the value. If return_header is True, also return the header.
 - b. get_column_index(header, column_name) returns the position of a column name in a header list. If the name is not in the header, return None. Python's list.index() does most of the work, but you need to handle the missing case. Test a name that is present, a name that is absent, and an empty header.
 - c. get_fire_gdp_year_data(co2_file, gdp_file, country) combines forest fire emissions and GDP for one country by matching years. Build it in these steps.
 - i. Use get_data to get the CO2 header and the CO2 rows for the country.
 - ii. Use get_data to get the GDP header and the GDP row for the country.
 - iii. Use get_column_index to find the Forest fires column in the CO2 header.
 - iv. For each CO2 row, use get_column_index to find that row's year in the GDP header, then take the value at that position in the country's GDP row.
 - v. Skip any year where the forest fire value or the GDP value is empty, or where the year is not in the GDP header.
 - vi. Return a list of [year, forest_fires, gdp] entries, with the year as an int and the other two as floats.
 - vii. Example. For Brazil in 2005, '2005' is at index 56 of the GDP header, and index 56 of Brazil's GDP row is 2170584.50.
5. Commit after each test and code cycle. Just commit, don't push. You do not need to push until you are finished.
6. Make a small scientific presentation on the relationship between forest fires and GDP for several countries. Look at each country on its own rather than pooling them, since GDP is in local currency. Write a few sentences in your README for each of the following.

- a. Introduction. What are you looking at?
 - b. Results. What did you find? Include your plots. You may base your plotting code on `src/scatter.py`. That script does not follow our best practices. Every file you submit must follow best practices.
 - c. Methods. What steps did you take to get your results? Include the commands someone would run to make your plots from the downloaded data.
7. New code must be properly tested. To test if a file is created in `sssh` test use the following assert

```
assert_equal $file_name $( ls $file_name )
```
 8. All unit and functional tests must be run and passing with github actions
 9. Create a release from main tagged as 1.0